

Structured Semiconductor Financial News and Volatility Dataset

for SMHandSOXX. (2020–2025)

A reusable research dataset combining 23,430 institutional news headlines with ETF market data, Garman–Klass volatility, FinBERT sentiment scores, and intraday microstructure variables

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Dataset Overview

The first openly documented dataset that aligns institutional semiconductor news with high-frequency ETF microstructure, ready for volatility forecasting and event studies.

23,430

2020–2025

1,508
TRADING DAYS

2
ETFs · SMH / SOXX

5
FORECAST
HORIZONS

Use Cases



Financial NLP research

Benchmarking FinBERT & LLM sentiment models



Volatility forecasting

HAR, GARCH, ML & deep-learning baselines



Event studies

Earnings, export-control & policy shocks



Semiconductor market research

SMH / SOXX, sector cycles, capex regimes



AI regime analysis

Pre / post-ChatGPT & export-control eras

Technical Stack

All processing in open-source Python; reproducible from the included Jupyter notebook.

Python 3.11

pandas

numpy

yfinance

HuggingFace Transformers

FinBERT

scikit-learn

statsmodels

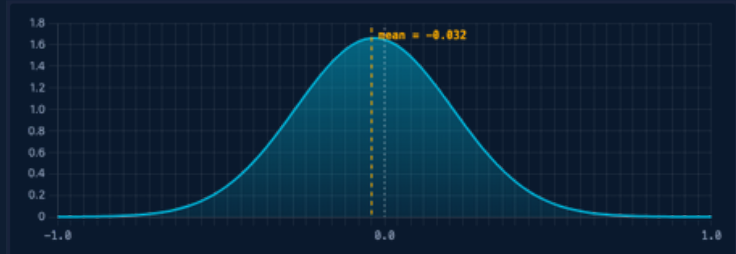
matplotlib · seaborn

Jupyter Notebook

Sentiment Distribution

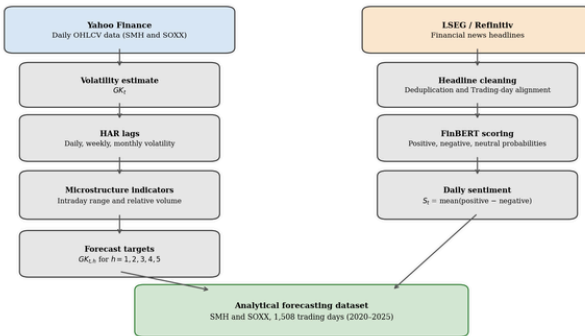
FIGURE 3.1 — MONTHLY NEWS COVERAGE (ATTACHED)

DAILY NET SENTIMENT DISTRIBUTION (S_t)

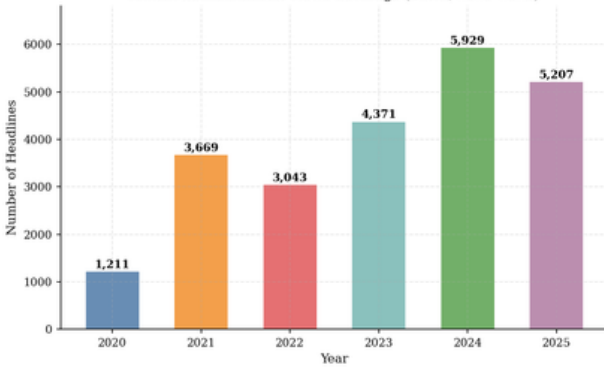


Daily net sentiment S_t is approximately normal with mean -0.032 — a mild negativity bias consistent with risk-coverage emphasis in institutional headlines.

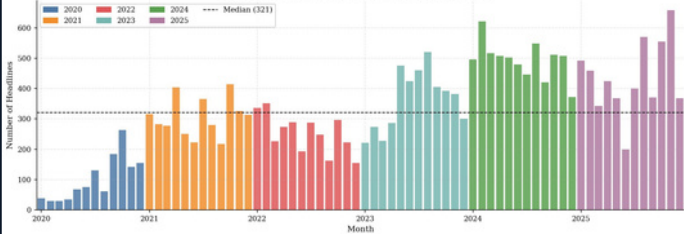
Data Pipeline and Analytical Sample Construction



Annual Semiconductor News Coverage (LSEG, 2020-2025)



Monthly Semiconductor News Coverage (LSEG, 2020-2025)



SMH — Garman-Klass Daily Volatility (Jan 2020 - Dec 2025)



SOXX — Garman-Klass Daily Volatility (Jan 2020 - Dec 2025)

